

Yasaman Saffari

AI Engineer — Dialogue Systems, LLMs, Speech & NLP

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About Me

I'm passionate about solving real-world problems using AI, particularly in dialogue systems, LLMs, speech processing, and reinforcement learning. I've contributed to several large-scale industrial projects, including an online speech-to-text system for banking applications, short-text language detection, a knowledge-graph feasibility study for enterprise cybersecurity, and a change-detection system for urban monitoring (Rafsanjan Municipality, grant-funded).

My academic background (Ph.D., AI, Kashan University) has given me a strong research foundation, but what I'm looking for now is hands-on industry experience, I'm open to starting at an entry level or internship position to build that experience. I enjoy diving into unfamiliar technical areas and learning quickly enough to deliver results on tight deadlines.

Skills:

- Dialogue Systems, Deep Reinforcement Learning (DRL), Large Language Models (LLMs), Natural Language Processing (NLP)
 - Graph Representation Learning, Knowledge Graphs, Data Analysis & Modeling, Clustering & Probabilistic Modeling
 - Machine Learning, Deep Learning, Computer Vision, Data Visualization & Statistical Analysis
 - Python, TensorFlow, PyTorch, Hugging Face Transformers, Stable-Baselines3, PyTorch Geometric, Scikit-learn
 - Weights & Biases, TensorBoard, Docker, Git, VS Code
 - Retrieval-Augmented Generation (RAG), LangChain, Vector Databases (Qdrant), Re-ranking, Prompt Engineering for LLM Applications, Hugging Face Transformers, RAGAS, Document Question Answering
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Projects

End-to-End Retrieval-Augmented Generation (RAG) System for Knowledge Retrieval from Unstructured Documents(2026) :Developed an end-to-end Retrieval-Augmented Generation (RAG)

system using LangChain, Qdrant, BGE embeddings, and Qwen. The system indexed PDF documents, performed semantic retrieval, generated grounded answers with source citations, and reduced hallucinations through document-based context retrieval.

Dialogue System (2019 – Present)

Ph.D. thesis at Kashan University (Supervisor: Dr. Salimi); Task-Oriented Dialogue Systems using Deep Reinforcement Learning, Graph Embedding, Attention Mechanism, and Probabilistic Topological State Representation with GMM.

Organizational Knowledge Graph Design and Feasibility Study for Cybersecurity (Fall 2025 – Winter 2025)

Non-academic project at Kashan University (Supervisors: Dr. Javad Salimi Sartakhti and Dr. Vahidipour); Knowledge Graphs, Cybersecurity, Organizational Knowledge Modeling and Feasibility Analysis.

Change Detection in Surveillance Images of Specialized Environments (2025)

Grant-winning non-academic project for Rafsanjan Municipality at Kashan University (Supervisor: Dr. Javad Salimi Sartakhti); Computer Vision, Image Processing, Change Detection, and Environmental Monitoring.

Language Detection in Multi-Language Short Utterances (Oct 2024 – Mar 2025)

Non-academic project at Kashan University (Supervisor: Dr. Salimi); LLM, FastText, BERT; Online Speech-to-Text (VAD).

Air Pollution Prediction (Apr 2023)

Non-academic project; Regression models (Linear, Gradient Boosting, KNN, Decision Tree, Random Forest); RNN and LSTM.

Fine-Tune XLSR Wav2Vec2 FA (Apr 2023)

Non-academic project at Kashan University (Supervisor: Dr. Salimi); Transformers for Online Speech-to-Text (VAD) in Farsi.

Spell Checker FA (May 2023)

Non-academic project at Kashan University; Fine-tuning bert-base-parsbert-uncased.

Car Plate Detection (Jan 2022 – July 2022)

Big project (Non-academic, Supervisor: Dr. Salimi); Yolo5; Car plate and character detection; Dataset labeling.

POS Tagging FA (Sep 2020 – Feb 2021)

Course project at Kashan University; Bi-LSTM-CNN model for Persian POS tagging.

ESS for Pool Population under BWA Attack (Feb 2021 – Aug 2021)

Course project at Kashan University (Game Theory); Python implementation.

Troublesome Travel Game (2020)

Global Game Jam 2020 (1st place, Sharif University of Technology); 3D Game in Unity, Role: Programmer (Atlas Team).

Voice Cloning of Voice Actors (Feb 2019 – Feb 2020)

Master's thesis at Tabriz Islamic Art University (Supervisor: Dr. Abbas Ghaffari); Voice Cloning using Deep Neural Networks for phonological modeling in dubbing.

Virtual Dance Training System (Sep 2018 – May 2019)

Exhibited at Tabriz Creativity & Innovation Expo; Virtual Reality environment + motion capture; Movement analysis using heuristic matching; Praised by university president.

Music Playlist Explorer Assistant (Sep 2018 – Feb 2019)

Course project at Tabriz Islamic Art University; Voice search and speech-to-text using Google TTS, implemented in Python.

Laboratory and Blood Bank Website (Sep 2016 – July 2017)

Bachelor thesis at Shariaty Technical and Vocational University; Website design using PHP (MVC, CodeIgniter).

Some Small Projects

- COVID-19 Prediction (June 2021 - Sep 2021): *RNN*, Neural Networks, Kashan University, Iran
- Face Mask Recognition (Feb 2021 - July 2021): *VGG, Hopfield*, Neural network, Kashan University, Iran

- Bert-fa-finetune (Jan 2021 - June 2021): *Non-academic*, Supervisor: Dr. Salimi, Persian-wikipedia-dataset, Kashan University, Iran
 - Handwritten Digit Recognition (Farsi) (Jan 2021 - June 2021): MLP, Hoda dataset, Pattern recognition, Kashan University, Iran
 - Face Recognition (Sep 2020 - Feb 2021): Paper implementation: Dissimilarity-based nearest neighbor classifier for single-sample face recognition (Zhengqi Zhang, Li Zhang, Meng Zhang), Pattern recognition, Kashan University, Iran
 - Simulation of Physics and Projectile Motion Formulas (Feb 2019 - Feb 2020): MATLAB image processing, Graphic Programming 1, MultiMedia Department, Tabriz Islamic Art University, Iran
 - Balance 2D Runner Game (2019): *Serious Game Prizes (Hacatoon)*, Leading the Telma+ Team, Unity, MultiMedia Department, Tabriz Islamic Art University, Iran
 - 3D Happy Pencil Game (Feb 2019): *Global Game Jam international event*, Game scoring system, Code and Play: [Happy Pencil](<https://v3.globalgamejam.org/2019/games/happypencil>), MultiMedia Department, Tabriz Islamic Art University, Iran
 - Tourism Data Visualization (Feb 2018 - Aug 2018): *Visualization of huge tourism data in the form of an application*, Processing software (Java), Visualization of Information, MultiMedia Department, Tabriz Islamic Art University, Iran
 - 2D Spider-Man Game (Aug 2017 - Aug 2017): Game Maker software, Personal
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Awards

- The 3rd place of Persian IIm-hackathon in "selection stage" (2024): *quera-[Results](<https://quera.org/contest/assignments/64356/scoreboard/>)*
 - Full fund with quota of brilliant talents (Sep 2020 - Present): *Kashan University, Iran - PhD position in AI, Computer Engineering*
 - The second rank of graduation of the master's degree (Feb 2020): *Tabriz Islamic Art University, Iran - MultiMedia Department*
 - The first rank of Global Game Jam 2020 (2020): *Sharif University Of Technology - Code and Play: [troublesome-travel](<https://v3.globalgamejam.org/2020/games/troublesome-travel-9>)*
 - Winning the second place in the egg throwing competition of Shariati College (2014): *Shariaty Technical and Vocational University*
 - Awarded a prestigious full-fund scholarship from Allameh Tabatabai elite pre-university (2012)
 - Obtaining the second place in the region in drawing with colored pencils (2011): *Zahra Mardani High School for Girls 2, Tehran, Iran*
 - Passing the first stage of the National Computer Olympiad (2010): *Zahra Mardani High School for Girls 2, Tehran, Iran*
 - Obtaining the first place in the Robocup competition of the 10th region of Tehran province (2010): *Zahra Mardani High School for Girls 2, Tehran, Iran*
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Education

Ph.D. in Artificial Intelligence (2020 – 2026)

Kashan University, Iran; Supervisor: Dr. Javad Salimi Sartakhti; Research on Dialogue Systems, Reinforcement Learning, State Representation Learning, Graph Representation Learning, Attention, and Transformers; GPA: 3.81. Thesies grade: A, 19.65/20

Master of Arts in Intelligent Simulator Design (2018 – 2020)

Tabriz Islamic Art University, Iran; Supervisor: Dr. Abbas Ghaffari; Second Instructor: Dr. Leila Dubakhti; Advisor: Dr. Alizadeh; Thesis: Investigating the Effective Variables of Speech Phonology at the Audience Origin Using Simulation of Lasting Voices in Dubbing (Voice Cloning with Deep Neural Networks); GPA: 3.79.

Bachelor of Software Computer Engineering (2013 – 2018)

Shariaty Technical and Vocational University, Iran; Supervisor: Mr. Yousef MusaZadeh; Thesis: Blood Laboratory Website Design using PHP; GPA: 3.46.

Publications

Published

1. Y Saffari, J Salimi Sartakhti, "A Graph-Based State Representation Learning In Episodic RL for Task-Oriented Dialogue Systems", *Engineering Applications of Artificial Intelligence*, 2025, Q1, IF=8
2. "CODACAN : Distributional Actor Critic with Hierarchical Attention-based State Representation for Dialogue Policy Learning", *Knowledge-Based Systems* , 2025, Q1, IF=7.6
3. Y Saffari, J Salimi Sartakhti, "Actor Double Critic Architecture for Dialogue System," *Journal of Electrical and Computer Engineering Innovations (JECEI)*, 11 (2), 2023
4. SA Faregh, Y Saffari, M Jafari Sisis, "The Impact of Gamification on the Teaching and Learning Process of Conflict Management," *Technology of Education Journal (TEJ)*, 15 (4), 657-672, 2021
5. Y Saffari, A Ghaffari, L Dobakhi, B Alizade-Ashrafi, "Computer Games for Dubbing Teaching: The 'Dublor Sho' Game," *The 5th International Conference on Computer Games; Challenges and Opportunities*, 2020
6. Y Saffari, S Nasre-Esfahani, R Hayati, Y Sekhavat, " Eng Title: Investigating the effectiveness of teaching based on movement imitation in a virtual reality environment for Azari dance," *The 4th International Conference on Computer Games; Challenges and Opportunities*, 2019

Not published (Y Saffari, J Salimi Sartakhti)

Lab and Visiting Experience

Acoustic Lab Researcher (Sep 2019 – Jan 2020)

MultiMedia Department, Tabriz Islamic Art University, Iran; Collaborated on establishing and managing the Acoustic Lab, including equipment setup and organization; Used lab resources for Master's thesis research.

VR Lab Development and Research (Sep 2018 – June 2019)

MultiMedia Department, Tabriz Islamic Art University, Iran; Founded and managed the VR Lab; Conducted research on VR gaming applications; Developed a "virtual dance teaching" system using VR and motion capture.

Visiting Student (Jun 2014 – Sep 2014)

Faculty of Computer Science and Engineering, Beheshti University, Tehran, Iran; Attended the Engineering Statistics course.

Scientific Talks

Reinforcement Learning for Task-Oriented Dialogue Systems (Feb 2024)

Cognitive Systems Research Group, University of Gothenburg

Presentation on RL-based dialogue systems, covering attentional and graph-based structural state representations.

Teaching Experience

- Invited Lecturer (Feb 2022 - Present): Simulation and Modeling of System 2 (Python and ML in Game and Art) (Master students), MultiMedia Dept., Tabriz Islamic Art Univ., Iran (2 semesters)

- Teaching Assistant (Sep 2023 - Present): Game Theory, Kashan University, Iran

- Invited Lecturer (Feb 2021 - Jun 2022): Sound Prod. & Design (Music - FL Studio) (Bachelor students), MultiMedia Dept., Tabriz Islamic Art Univ., Iran (3 semesters)

- Teaching Assistant (Feb 2019 - Jun 2019): Simulation and Modeling of System 1 (VR lab) (Master students), MultiMedia Dept., Tabriz Islamic Art Univ., Iran

- Teaching Assistant (Sep 2015 - Feb 2016): Microprocessors and Assembly Language (Dr. Hassan Seyed Razi) Book: AVR microcontrollers by Dr. Hassan Seyed Razi, Kian Univ. Press, Shariaty Technical and Vocational Univ., Iran

- Teaching Assistant (Sep 2014 - Feb 2015): Electric Circuits Book: Fundamental Theory of Circuits and Networks, Vol. 1 (Ernest Cough - Charles Deso), Shariaty Technical and Vocational Univ., Iran
